Pranav M

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Research Interests: Processing-Near-Memory / Near-Memory Computing, Memory System Architecture, DDR/Memory Interfaces, CPU & SoC Design

Education

2023 - 2027 B.Tech, Electronics and Communication Engineering, **PES University**
DAC (Distinction) Scholarship — merit-based, for academic excellence, Semester 5
2022 Class 12th, **Vydehi School of Excellence**

Research Experience

Final Year Project — Reconfigurable Universal DDR5 Memory Controller 2026 – Present
CHIPS Lab, PES University
Designing a reconfigurable, universal DDR5 memory controller with an AXI front-end, aimed at supporting multiple memory configurations through a single reusable controller architecture.

Aletheia — Processing-Near-Memory Framework 2026
Independent Research Project GitHub
Built an experimental processing-near-memory (PNM) framework on a Raspberry Pi 5, dispatching commands over TCP to the Pi as a memory node instead of moving data. Ran PMU-backed characterization of memory access patterns, including working-set sweeps, stride scans, and dataset-scaling experiments to study locality and cache-miss behavior.

Projects

Herald GitHub
Designed and taped out a fixed-point (Q12.12) DSP coprocessor for FPU-less microcontrollers via Tiny Tapeout (IHP SG13G2 130nm shuttle). Features a CORDIC engine (simultaneous sin/cos, atan2, vector magnitude, normalization) and a MAC unit (multiply, multiply-accumulate, multiply-subtract), verified in Bluespec and cocotb. Signed off clean at 50 MHz with roughly 6 ns worst-case setup slack, zero routing overflow, and ˜80k μm^2 area (62.9% utilization).

Tiny-Tone GitHub
Designed a PWM-based audio tone generator peripheral in Bluespec, submitted to and accepted into TinyQV, an open community peripheral contest built around a shared RISC-V SoC on Tiny Tapeout. The peripheral exposes a memory-mapped register interface for enabling/disabling the tone, setting frequency at runtime, and reading back status. Converting the Bluespec-generated Verilog to fit the TinyQV SoC interface required hand-mapping ports and building a wrapper; a cocotb-based debugging session traced a register readback failure to a byte-write vs. word-write mismatch between the peripheral and testbench.

Aetheron GitHub
Built a simulation-based RISC-V SoC from scratch in Bluespec SystemVerilog, running real compiled C programs. Includes a pipelined RV32I core, boot-from-ROM loading of C payloads into RAM, a TileLink-lite interconnect, memory-mapped UART/GPIO/Timer peripherals, and a full C/ASM toolchain with a custom linker script.

Chorion GitHub
Implemented the ChaCha20 stream cipher (RFC 8439 compliant, 20-round) as a synthesizable SystemVerilog core on the PYNQ-Z2 FPGA (Zynq-7000 SoC), validated against a Python reference implementation with a 100% match on RFC 8439 test vectors. Synthesized to 2,118 LUTs (10%) and 1,807 FFs (4%) on an Artix-7-equivalent fabric.

Mentorship & Outreach

HSP, PES University — Mentor Jul 2025 – Present
Mentored students on digital hardware design projects across two consecutive terms, guiding architecture design, reviewing HDL, and helping debug design modules.

Skills

Languages	SystemVerilog, Verilog, Bluespec, C, Python, Assembly
Tools & Platforms	Vivado, GTKWave, Vitis IDE, Verilator, ModelSim, Git, VS Code, Neovim, Linux
Domains	Digital Design, Computer Architecture, Memory Systems, SoC Design, RTL Simulation and Synthesis

Writing

Maintains a research-focused blog and weekly TIL/research-diary practice at prawns.dev, documenting ongoing work in computer architecture and memory systems.